

Problem 1

Part a)

The measurable space (common to both sub-problems) is given by $(\Omega, \mathcal{F})$, where: $\Omega = \mathbb{N}$ and $\mathcal{F} = \mathcal{P}(\mathbb{N})$.

i) $p_1(k) = Ck^{-2}$

Let $\mathbb{P}_1(F) := \sum_{k \in F} p_1(k) = \sum_{k \in F} Ck^{-2}, \forall F \in \mathcal{F}$ be a set function on the measurable space $(\mathbb{N}, \mathcal{P}(\mathbb{N}))$. To form a valid probability mass function, We need to find the value of C such that $p_1(k) \geq 0$ for all $k \in \mathbb{N}$ and $\sum_{k=1}^{\infty} p_1(k) = 1$. We find C by solving the equation:

$$\sum_{k=1}^{\infty} Ck^{-2} = C \sum_{k=1}^{\infty} \frac{1}{k^2} = 1$$

Using the limit $\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$, we obtain:

$$C \left(\frac{\pi^2}{6} \right) = 1 \implies C = \frac{6}{\pi^2}$$

Since $C > 0$, $p_1(k) \geq 0$ holds for all $k \in \mathbb{N}$. Therefore, $p_1(k) = \frac{6}{\pi^2 k^2}$ is a valid probability mass function and $(\mathbb{N}, \mathcal{P}(\mathbb{N}), \mathbb{P}_1)$ is a probability space, with $\mathbb{P}_1$:

$$\mathbb{P}_1(F) = \sum_{k \in F} p_1(k) = \sum_{k \in F} \frac{6}{\pi^2 k^2} = \frac{6}{\pi^2} \sum_{k \in F} \frac{1}{k^2}, \forall F \in \mathcal{F}$$

We evaluate $\mathbb{P}_1((1, 5])$ and $\mathbb{P}_1(2\mathbb{N})$:

- $\mathbb{P}_1((1, 5])$: Since the probability measure $\mathbb{P}_1$ is defined strictly on $\mathbb{N}$, evaluating the continuous interval $(1, 5]$ requires defining the event as its intersection with the sample space:

$$A := (1, 5] \cap \mathbb{N} = \{2, 3, 4, 5\}$$

Then:

$$\begin{aligned} \mathbb{P}_1((1, 5]) &= \mathbb{P}_1(A) = \mathbb{P}_1(\{2, 3, 4, 5\}) \\ &= \frac{6}{\pi^2} \sum_{k=2}^5 \frac{1}{k^2} = \frac{6}{\pi^2} \left(\frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \frac{1}{25} \right) \\ &= \frac{6}{\pi^2} \left(\frac{900 + 400 + 225 + 144}{3600} \right) = \frac{6}{\pi^2} \frac{1669}{3600} = \frac{1669}{600\pi^2} \end{aligned}$$

- $\mathbb{P}_1(2\mathbb{N})$: Let B be the event containing all even natural numbers, i.e., $B := 2\mathbb{N} = \{2k \mid k \in \mathbb{N}\}$:

$$\begin{aligned} \mathbb{P}_1(2\mathbb{N}) &= \sum_{k=1}^{\infty} p_1(2k) = \frac{6}{\pi^2} \sum_{k=1}^{\infty} \frac{1}{(2k)^2} = \frac{6}{4\pi^2} \sum_{k=1}^{\infty} \frac{1}{k^2} \\ &= \frac{6}{4\pi^2} \cdot \frac{\pi^2}{6} \quad \left(\text{since } \sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6} \right) \\ &= \frac{1}{4} \end{aligned}$$

ii) $p_2(k) = C \frac{2^k}{k!}$

Let $\mathbb{P}_2(F) := \sum_{k \in F} p_2(k) = \sum_{k \in F} C \frac{2^k}{k!}$, $\forall F \in \mathcal{F}$ be a set function on the measurable space $(\mathbb{N}, \mathcal{P}(\mathbb{N}))$. To form a valid probability mass function, we need to find the value of C such that $p_2(k) \geq 0$ for all $k \in \mathbb{N}$ and $\sum_{k=1}^{\infty} p_2(k) = 1$. We find C by solving the equation:

$$\sum_{k=1}^{\infty} p_2(k) = C \sum_{k=1}^{\infty} \frac{2^k}{k!} = 1$$

Using the standard Taylor expansion of e^x evaluated at $x = 2$:

$$\sum_{k=1}^{\infty} \frac{2^k}{k!} = \left(\sum_{k=0}^{\infty} \frac{2^k}{k!} \right) - \frac{2^0}{0!} = e^2 - 1$$

Thus:

$$C(e^2 - 1) = 1 \implies C = \frac{1}{e^2 - 1}$$

Since $C > 0$ and $e^2 > 1$, $p_2(k) \geq 0$ holds for all $k \in \mathbb{N}$. Therefore, $p_2(k) = \frac{1}{e^2 - 1} \frac{2^k}{k!}$ is a valid probability mass function and $(\mathbb{N}, \mathcal{P}(\mathbb{N}), \mathbb{P}_2)$ is a probability space, with $\mathbb{P}_2$:

$$\mathbb{P}_2(F) = \sum_{k \in F} p_2(k) = \sum_{k \in F} \frac{1}{e^2 - 1} \frac{2^k}{k!} = \frac{1}{e^2 - 1} \sum_{k \in F} \frac{2^k}{k!}, \forall F \in \mathcal{F}$$

We evaluate $\mathbb{P}_2((1, 5])$ and $\mathbb{P}_2(2\mathbb{N})$:

- $\mathbb{P}_2((1, 5])$: Using the previously defined event $A := (1, 5] \cap \mathbb{N} = \{2, 3, 4, 5\}$:

$$\begin{aligned} \mathbb{P}_2((1, 5]) &= \mathbb{P}_2(A) = \sum_{k=2}^5 p_2(k) = \frac{1}{e^2 - 1} \left(\frac{2^2}{2!} + \frac{2^3}{3!} + \frac{2^4}{4!} + \frac{2^5}{5!} \right) \\ &= \frac{1}{e^2 - 1} \left(2 + \frac{8}{6} + \frac{16}{24} + \frac{32}{120} \right) = \frac{1}{e^2 - 1} \left(2 + \frac{4}{3} + \frac{2}{3} + \frac{4}{15} \right) \\ &= \frac{64}{15(e^2 - 1)} \end{aligned}$$

- $\mathbb{P}_2(2\mathbb{N})$: Using the previously defined event $B := 2\mathbb{N} = \{2k \mid k \in \mathbb{N}\}$:

$$\begin{aligned} \mathbb{P}_2(2\mathbb{N}) &= \sum_{k=1}^{\infty} p_2(2k) = \frac{1}{e^2 - 1} \sum_{k=1}^{\infty} \frac{2^{2k}}{(2k)!} \\ &= \frac{1}{e^2 - 1} (\cosh(2) - 1) \quad \left(\text{since } \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!} = \cosh(x) \right) \\ &= \frac{1}{e^2 - 1} \left(\frac{e^2 + e^{-2}}{2} - 1 \right) = \frac{1}{e^2 - 1} \left(\frac{e^4 - 2e^2 + 1}{2e^2} \right) \\ &= \frac{1}{e^2 - 1} \cdot \frac{(e^2 - 1)^2}{2e^2} = \frac{e^2 - 1}{2e^2} \end{aligned}$$

Part b)

i) Binomial and Poisson distributions

Poisson Distribution: The probability mass function is given by $p(k) = \frac{\lambda^k}{k!}e^{-\lambda}$ for $k \in \mathbb{N}_0$ and $\lambda > 0$. We want to show that $p(k-1)p(k+1) \leq p(k)^2$ for all $k \geq 1$.

$$\begin{aligned} & \frac{\lambda^{k-1}}{(k-1)!}e^{-\lambda} \frac{\lambda^{k+1}}{(k+1)!}e^{-\lambda} \leq \left(\frac{\lambda^k}{k!}e^{-\lambda} \right)^2 \\ \iff & \frac{\lambda^{k-1+k+1}}{(k-1)!(k+1)!}e^{-2\lambda} \leq \frac{\lambda^{2k}}{k! \cdot k!}e^{-2\lambda} \\ \iff & \frac{\lambda^{2k}e^{-2\lambda}}{(k-1)!(k+1)k!} \leq \frac{\lambda^{2k}e^{-2\lambda}}{k(k-1)! \cdot k!} \end{aligned}$$

Since $\lambda > 0$ and $k \geq 1$, the exponential, power, and factorial terms are strictly positive, then the common terms can be canceled from both sides:

$$\iff \frac{1}{k+1} \leq \frac{1}{k} \iff k \leq k+1 \iff 0 \leq 1$$

Because the sequence is built on logical equivalences ($\iff$) and concludes with a trivially true statement, the initial inequality holds for all $k \in \mathbb{N}$. □

Binomial Distribution: The probability mass function is given by $p(k) = \binom{n}{k}p^k(1-p)^{n-k}$ for $k \in \{0, \dots, n\}$ and $p \in [0, 1]$. We want to show that $p(k-1)p(k+1) \leq p(k)^2$ for all $k \geq 1$. For the boundary cases where $p \in \{0, 1\}$, $p(k-1)p(k+1) = 0$ for all $k \geq 1$, which trivially satisfies $0 \leq p(k)^2$. For $k \geq n$, $p(k+1) = 0$, making the left side of the inequality 0, which also trivially satisfies $0 \leq p(k)^2$. For $1 \leq k \leq n-1$ and $p \in (0, 1)$:

$$\begin{aligned} & \binom{n}{k-1}p^{k-1}(1-p)^{n-k+1} \binom{n}{k+1}p^{k+1}(1-p)^{n-k-1} \leq \left(\binom{n}{k}p^k(1-p)^{n-k} \right)^2 \\ \iff & \binom{n}{k-1} \binom{n}{k+1} p^{2k}(1-p)^{2(n-k)} \leq \binom{n}{k}^2 p^{2k}(1-p)^{2(n-k)} \end{aligned}$$

Since $p \in (0, 1)$, both $p > 0$ and $(1-p) > 0$, we can cancel these terms from both sides:

$$\begin{aligned} \iff & \binom{n}{k-1} \binom{n}{k+1} \leq \binom{n}{k}^2 \\ \iff & \frac{n!}{(k-1)!(n-k+1)!} \cdot \frac{n!}{(k+1)!(n-k-1)!} \leq \frac{(n!)^2}{(k!)^2((n-k)!)^2} \\ \iff & \frac{1}{(k-1)!(n-k+1)(n-k)!(k+1)k!(n-k-1)!} \leq \frac{1}{k(k-1)!k!(n-k)(n-k-1)!(n-k)!} \end{aligned}$$

Since $1 \leq k \leq n-1$, all factorial terms are strictly positive and can be canceled from both sides:

$$\begin{aligned} &\iff \frac{1}{(n-k+1)(k+1)} \leq \frac{1}{k(n-k)} \\ &\iff \frac{k}{k+1} \cdot \frac{n-k}{n-k+1} \leq 1 \end{aligned}$$

For $1 \leq k \leq n-1$, both $k > 0$ and $n-k > 0$. Therefore, the fractions satisfy $\frac{k}{k+1} < 1$ and $\frac{n-k}{n-k+1} < 1$. The product of two positive numbers strictly less than 1 is strictly less than 1, meaning the inequality holds. $\square$

ii) $p(k) = 90/(\pi k)^4$

The probability mass function is given by $p(k) = \frac{90}{(\pi k)^4}$ for $k \in \mathbb{N}$. We want to show that $p(k-1)p(k+1) \geq p(k)^2$ for $k \geq 2$.

$$\begin{aligned} &\frac{90}{(\pi(k-1))^4} \cdot \frac{90}{(\pi(k+1))^4} \geq \left(\frac{90}{(\pi k)^4} \right)^2 \\ &\iff \frac{1}{(k-1)^4(k+1)^4} \geq \frac{1}{k^4 \cdot k^4} \quad (\text{canceling positive constants}) \\ &\iff k^4 \cdot k^4 \geq (k-1)^4(k+1)^4 \quad (\text{since } k \geq 2) \end{aligned}$$

Since $f(x) = \sqrt[4]{x}$ is a monotonically increasing function for $x > 0$, the inequality direction is preserved when taking the fourth root of both sides:

$$\iff k \cdot k \geq (k-1)(k+1) \iff k^2 \geq k^2 - 1 \iff 0 \geq -1$$

This final statement is universally true. Since all algebraic steps performed are reversible, the original inequality holds for all $k \geq 2$. $\square$

iii) **Existence of PMF satisfying** $p(k)^2 = p(k-1)p(k+1)$

We divide the problem into two mutually exclusive cases:

Case 1: Assuming $p(k) > 0$ **for all** $k \in \mathbb{N}_0$:

$$\begin{aligned} &p(k)^2 = p(k-1)p(k+1) \\ &\iff \frac{p(k)}{p(k-1)} = \frac{p(k+1)}{p(k)} := c \quad \forall k \geq 1 \end{aligned}$$

where $c > 0$ is a constant, i.e., the ratio between any two consecutive terms is always c :

- For $k = 1$: $\frac{p(1)}{p(0)} = c \implies p(1) = cp(0)$
- For $k = 2$: $\frac{p(2)}{p(1)} = c \implies p(2) = cp(1) = c^2p(0)$

Assume it holds for $n \in \mathbb{N}$, i.e., $p(n) = c^n p(0)$. For $k = n + 1$:

$$\frac{p(n+1)}{p(n)} = c \implies p(n+1) = cp(n) = c(c^n p(0)) = c^{n+1} p(0)$$

By induction, $p(k) = p(0) \cdot c^k$. By the fundamental axiom of probability, the sum over the sample space $\mathbb{N}_0$ must equal to 1 to form a valid probability mass function:

$$\sum_{k=0}^{\infty} p(k) = 1 \implies \sum_{k=0}^{\infty} p(0)c^k = 1 \implies p(0) \sum_{k=0}^{\infty} c^k = 1$$

Given that $p(0) \in (0, 1)$, for this series to converge to a finite constant, the common ratio must satisfy $0 < c < 1$. Using the standard geometric series formula $\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}$:

$$p(0) \left(\frac{1}{1-c} \right) = 1 \iff p(0) = 1 - c$$

Plugging $p(0) = 1 - c$ back into our sequence $p(k) = p(0)c^k$, we obtain $p(k) = (1 - c)c^k$. Defining $q := 1 - c$ (which implies $c = 1 - q$), we get:

$$p(k) = q(1 - q)^k$$

This formula corresponds to the probability mass function of the geometric distribution, which fulfills the condition for $p(k) > 0$.

Case 2: There exists at least one $m \in \mathbb{N}_0$ such that $p(m) = 0$:

Let us analyze how a zero value propagates through the recurrence relation $p(k)^2 = p(k-1)p(k+1)$, which is valid for all $k \geq 1$:

- **Forward Propagation:** Suppose $p(m) = 0$ for some index $m \in \mathbb{N}_0$. Evaluating the recurrence at the next index $k = m + 1$ yields:

$$p(m+1)^2 = p(m)p(m+2) = 0 \cdot p(m+2) = 0 \implies p(m+1) = 0$$

By induction, if any term is zero, all subsequent terms must be zero (i.e., $p(k) = 0, \forall k \geq m$).

- **Backward Propagation:** Suppose $p(m) = 0$ for some index $m \geq 2$. Evaluating the recurrence at the preceding index $k = m - 1$ yields:

$$p(m-1)^2 = p(m-2)p(m) = p(m-2) \cdot 0 = 0 \implies p(m-1) = 0$$

By induction, all preceding terms down to index 1 must also be zero (i.e., $p(k) = 0, \forall 1 \leq k \leq m$).

Therefore, the entire sequence from index 1 onward must be zero: $p(k) = 0, \forall k \geq 1$

At the lowest valid index ($k = 1$), we have:

$$p(1)^2 = p(0)p(2) \implies 0^2 = p(0) \cdot 0 \implies 0 = 0$$

which holds true for any value of $p(0)$ and, by the fundamental axiom of probability, we have:

$$\sum_{k=0}^{\infty} p(k) = p(0) + \sum_{k=1}^{\infty} 0 = 1 \implies p(0) = 1$$

This yields the sequence $(1, 0, 0, \dots)$, which corresponds exactly to the geometric distribution $g(q; k) = q(1 - q)^k$ evaluated at $q = 1$.

Final answer: The condition is satisfied exactly by the geometric distribution for parameter $q \in (0, 1]$. The probability space is given by $(\Omega, \mathcal{F}, \mathbb{P})$ where $\Omega = \mathbb{N}_0$, $\mathcal{F} = \mathcal{P}(\mathbb{N}_0)$, and $\mathbb{P}(A) = \sum_{k \in A} q(1 - q)^k, \forall A \in \mathcal{F}$, with $q \in (0, 1]$.